NSW INDEPENDENT TRIAL EXAMS – 2023
MATHEMATICS STANDARD YEAR 11 EXAMINATION
MARKING GUIDELINES

Section I

Question	Answer	Assessed Outcome	Band
1	D	MS-S1.2/MS11-7	3
2	B	MS-F1.2/MS11-6	3
3	B	MS-M2/MS11-3	3
4	C	MS-A2/MS11.2	3
5	A	MS-S2/MS11-8	4
6	D	MS-M1.2/MS11-4	4
7	C	MS-S1.1/MS11-7	3
8	D	MS-S1/MS11-7	4
9	D	MS-A1/MS11-10	4
10	C	MS-A4/MS11-9	4
11	C	MS-M1.2/MS11-4	4
12	D	MS-F1.3/MS11-2, MS11-6	4
13	D	MS-S1.2/MS11-2	4
14	C	MS-A1/MS11-1	5
15	A	MS-M1.1/MS11-3	5
16	C	MS-M1.2/MS11-3	5
17	B	MS-M1.2/MS11-4	5
18	C	MS-S1.1/MS11-7	5
19	C	MS-M2/MS11-10	5
20	A	MS-A1/MS11-6	6

Section II

<table>
<tr><th>Qu</th><th>Answer</th><th>Mark</th><th>Outcome Assessed</th><th>Band</th></tr>
<tr><td>21</td><td>Let $x = 2$, then $3^x + 3^{-x}$
$= 3^2 + 3^{-2}$
$= 9 + \frac{1}{9}$
$\mathbf{= 9\frac{1}{9}}$ or 9.1</td><td>1</td><td>MS-A1/
MS11-9</td><td>3</td></tr>
<tr><td>22</td><td>Let r = 5.5 ÷ 2 = 2.75

Then $V = \frac{4}{3} \times \pi \times 2.75^3$

$\mathbf{= 87.1\ cm^3}$</td><td>1

1</td><td>MS-A1/
MS11-9

MS-M1.2/
MS11-3</td><td>3</td></tr>
<tr><td>23</td><td>$-4m = 9 - m$
$-3m = 9$
$\mathbf{m = -3}$</td><td>
1
1</td><td>MS-A1/
MS11-10</td><td>3</td></tr>
<tr><td>24</td><td>\$US 850 = \$AUS $\frac{1}{0.764} \times 850$

= \$AUS 1112.57</td><td>

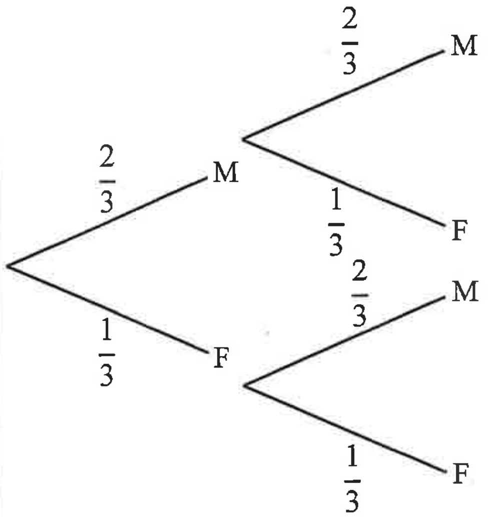
1</td><td>MS-A2/
MS11-6</td><td>3</td></tr>
<tr><td>25</td><td>$\sqrt{\frac{4\pi^3}{5}} = \sqrt{24.80502}$
= 4.98 (correct to 2 decimal places)</td><td>1

1</td><td>MS-A1/
MS11-6</td><td>3</td></tr>
<tr><td>26(a)</td><td><table>
<tr><th>Score</th><th>Frequency</th><th>Cumulative Frequency</th><th>Frequency × Score</th></tr>
<tr><td>15</td><td>7</td><td>7</td><td>105</td></tr>
<tr><td>25</td><td>3</td><td>10</td><td>75</td></tr>
<tr><td>35</td><td>8</td><td>18</td><td>280</td></tr>
<tr><td>45</td><td>16</td><td>34</td><td>720</td></tr>
<tr><td>55</td><td>6</td><td>40</td><td>330</td></tr>
<tr><td></td><td>40</td><td></td><td>1510</td></tr>
</table>Adding a frequency and frequency × score column, the mode is 45 with a frequency of 16.</td><td>1</td><td>MS-S1.1/
MS11-9</td><td>4</td></tr>
<tr><td>(b)</td><td>The mean = 1510 ÷ 40
= 37.75</td><td>1
1</td><td>MS-S1.1/
MS11-9</td><td>4</td></tr>
<tr><td>27</td><td>$2\pi r = 2.13 \times 10^4$
$r = (2.13 \times 10^4) \div 2\pi$
$= 3390$
Surface area of Mars = $4 \times \pi \times (3390)^2$ sq. km
$\mathbf{= 1.44 \times 10^8}$ sq. km</td><td>

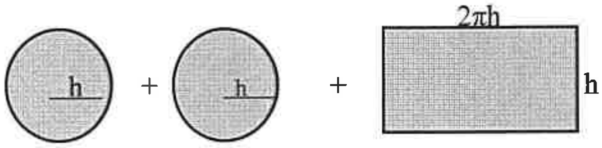
1
1
1</td><td>MS-M1.1/1.2
MS11-4</td><td>4</td></tr>
</table>

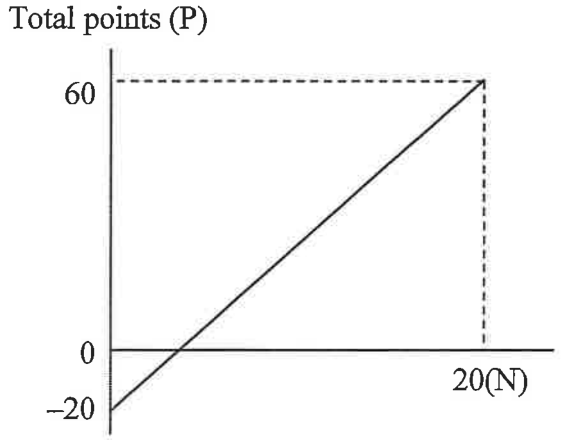
Qu	Answer	Mark	Outcome Assessed	Band
28(a)	**The range is 45 – 25 = 20**	1	MS-S1.2/ MS11-2	3
(b)	**The median is 35**	1	MS-S1.2/ MS11-2	3
(c)	Lowest Extreme = 25, Lower Quartile = 30, Median = 35, Upper Quartile = 40, Highest Extreme is 45. **5-number summary: 25, 30, 35, 40, 45**	1 1	MS-S1.2/ MS11-12	4
(d)	**The distribution is symmetrical** (there is no skewness and the median is in the centre of the distribution.)	1 1	MS-S1.2/ MS11-10	4
29(a)	**18–25 age group**	1	MS-S1/ MS11-2	3
(b)	In the 41–55 age group the percentage of residents who have been vaccinated $= \frac{25\ 000}{30\ 000} \times 100\%$ $= \mathbf{83.3\%}$	1	MS-M1.1/ MS11-3	3
(c)	**As the age of the residents increase, the percentage of these residents who have been vaccinated has increased.**	1	MS-S1.2/ MS11-10	3
(d)	P(at least one resident vaccinated) = 1 – P(neither of the residents have been vaccinated) $= 1 - \left(\frac{10\ 000}{15\ 000} \times \frac{10\ 000}{15\ 000}\right)$ $= 1 - \left(\frac{2}{3} \times \frac{2}{3}\right)$ $= \mathbf{\frac{5}{9}}$ **or 0.56**	1 1	MS-S2/ MS11-8	4
30	The length of the string is $4\pi r = 48$ cm The radius of the string is $48 \div 4\pi$ = 3.82 cm Length of cardboard = 8 × 3.82 = 30.56 cm Width of cardboard = 2 × 3.82 cm = 7.64 cm Perimeter of cardboard = (2 × 30.56) + (2 × 7.64) = **76.40 cm**	1 1 1	MS-M1.2/ MS11-4	4/5

Qu	Answer	Mark	Outcome Assessed	Band
31	The percentage of light passing through the two layers of glass is 88% × 88% $= 0.88 \times 0.88$ $= 0.7744$ $= 77.44\%$ **Loss of light is 22.56%**	 1 1	MS-F1.1/ MS11-6	5
32	Area of end $= \frac{0.35}{2}(0.5 + 0.3) + \frac{0.35}{2}(0.3 + 0)\ m^2$ $= 0.175 \times (0.8 + 0.3)\ m^2$ $= \mathbf{0.1925\ m^2}$ Volume of foam $= 0.1925 \times 1.2$ $= \mathbf{0.23\ m^3}$	1 1 1	MS-M1.2/ MS11-4	4/5
33	Interest earned over 5 years = 3.25 × 5 = 16.25% So, 116.25% = \$14 531.25 100% = \$14 531.25 ÷ 116.25 × 100 = \$12 500 **The amount over \$10 000 invested = \$2 500**	1 1 1	MS-F1.1/ MS11-9	5
34	Percent for "No Stamp" is 72.5 – 40.8 = **31.7** Cum % for "Postcode Unreadable" is 90.0 + 5.8 = **95.8**	1 1	MS-S1.1/ MS11-7	4

Qu	Answer	Mark	Outcome Assessed	Band
35(a)	**The ratio of Male:Female is 2:1** So, the probability of a Male baby is $\frac{2}{2+1}=\frac{2}{3}$	1	MS-S2/ MS11-10	4
(b)(i)		2	MS-S2/ MS11-9	4
(ii)	P(a female and a male baby) = P(M, F) or P(F, M) $=\left(\frac{2}{3}\times\frac{1}{3}\right)+\left(\frac{1}{3}\times\frac{2}{3}\right)$ $=\mathbf{\frac{4}{9}}$	1	MS-S2/ MS11-8	4
(iii)	From (ii), the probability that the animal will have a male and a female baby is $\frac{4}{9}$ P(both babies of the same sex) = P(M, M) or P(F, F) $=(\frac{2}{3}\times\frac{2}{3})+(\frac{1}{3}\times\frac{1}{3})$ $=\frac{5}{9}$ **So, it is more likely that the animal will have babies of the same sex.**	1	MS-S2/ MS11-10	4/5

Qu	Answer	Mark	Outcome Assessed	Band
36(a)	The perimeter of the pentagon is 88 cm So, $2x + 58 = 88$ $2x = 30$ $\boldsymbol{x = 15}$	 1 1	MS-M1.2/ MS11-4	4
(b)	Area of pentagon = area of rectangle + area of Δ Area of rectangle BCDE $= 18\text{ cm} \times 15\text{ cm}$ $= 270\text{ cm}^2$ Height of triangle $\sqrt{20^2 - 9^2}$ $= 17.86\text{ cm}$ Area of triangle $= \frac{1}{2} \times 18\text{ cm} \times 17.86\text{ cm}$ $= 160.75\text{ cm}^2$ Area of pentagon $= 270\text{ cm}^2 + 160.75\text{ cm}^2$ $\mathbf{= 430.75\text{ cm}^2}$	 1 1 1	MS-M1.2/ MS11-4	4/5
37	The amount of \$10 394.24 includes 17.5% for Jayden's holiday loading, so 117.5% = \$10 394.24 Jayden's salary for 4 weeks without the holiday loading = \$10 394.24 ÷ 117.5 × 100 = \$8846.16 **Annual salary = \$8846.16 ÷ 4 × 52** **= \$115 000.10**	1 1 1	MS-F1.2/ MS11-10	5

Qu	Answer	Mark	Outcome Assessed	Band
38(a)	Total surface area will be the area of two circles and a rectangle:  **Surface area S $= \pi h^2 + \pi h^2 + 2\pi h^2$** **$= 6\pi h^2$**	 2 1	MS-M1.2/ MS11-4 MS-A1/ MS11-10	5/6
(b)	If $S = 96\pi$ then $6\pi h^2 = 96\pi$ $h^2 = 16$ **$h = 4$** Now volume $= \pi \times h^2 \times 2h$ $= 2\pi h^3$ Substituting $h = 4$, $V =$ **$2 \times \pi \times 4^3$** **$= 402$ cm^2**	 1 1 1 1	MS-M1.2/ MS11-10	5
39(a)	The machine has lost \$120 000 – \$80 000 over a period of 8 years (i.e., \$5000/year). Annual rate of depreciation $= \dfrac{5000}{120\ 000} \times 100\%$ **$= 4.2\%$**	1 1	MS-F1.1/ MS11-9	5
(b)	The machine has already lost \$40 000 and is losing \$5000 each year, so **after another 2 years, the machine would have lost \$50 000.**	1	MS-F1.1/ MS11-10	4
(c)	The equation of the line is $V = 120\ 000 - 5000n$ where V is the value of the machine after n years. Let $120\ 000 - 5000n = 0$ $5000n = 120\ 000$ $n = 24$ **So, the machine will have no value after 24 years or after another 16 years.** **Alternate solution**: The machine is losing \$5000 each year. So, \$80 000 ÷ \$5000 = 16. So, the machine will have no value after another 16 years (i.e., a total of 24 years since new).	 1 1 (2)	MS-F1.1/ MS11-10	4/5

Qu	Answer	Mark	Outcome Assessed	Band
40(a)	**Maximum number of points = 20 × 3 = 60** **Minimum number of points = 20 × –1 = –20**	1 1	MS-A1/ MS-11-6	3/4
(b)		2	MS-A2/ MS11-1	5/6
(c)	The gradient of the graph is: 80/20 = 4, and the intercept on the vertical axis is –20. So, the equation is P = 4N – 20 **Let 4N – 20 = 0, then N = 5** **So, after 5 successful attempts (and 15 unsuccessful attempts) the player would score zero points.**	 1 1	MS-A2/ MS11-10	5/6

Qu	Answer	Mark	Outcome Assessed	Band
41(a)	From calculation, the mean of the set is 50.11 The sample standard deviation is 11.57 Now: Mean – (3 × sample standard deviation) = 50.11 – (3 × 11.57) = 15.4 Mean + (3 × sample standard deviation) = 50.11 + (3 × 11.57) = 84.82 **Since the data score of 25 lies within this range, it would not be considered an outlier using this method.**	1 1	MS-S1.2/ MS11-10	5/6
(b)	For this data set, the median is 52, $Q_1 = 45.5$, $Q_3 = 59$ and the IQR= 59 – 45.5 = 13.5 Consider $Q_1 - (1.5 \times \text{IQR})$ = 45.5 – (1.5 × 13.5) = 25.25 **Since the data score of 25 is just outside the lower limit using this method, it could possibly be considered an outlier for this data set.**	1 1	MS-S1.2/ MS11-10	5
(c)	The sample standard deviation gives the spread of scores from the mean and in this case, are quite widely spread, while the second method gives a more consistent result by using just the lower 25% of the data scores. **The second in this case would be a more reliable result.**	1 1	MS-S1.2/ MS11-10	6

Qu	Answer	Mark	Outcome Assessed	Band
42	D represents 110% of the invoice. So, the amount on the invoice before the GST was added = D ÷ 110 × 100 $= \frac{100D}{110}$ $= \mathbf{\frac{10D}{11}}$	1 1	MS-F1.1/ MS11-10	5/6
43	In order to have tax payable of \$21 342, Jason's taxable income must be in the range: \$45 001 – \$120 000 Now, \$21 342 – \$5092 = \$16 250 \$16 250 ÷ 0.325 = \$50 000 **So, Jason's taxable income was:** **\$45 000 + \$50 000** **= \$95 000**	1 1 1	MS-F1.2/ MS11-9	6